

Pillar 1 · advantage estimator

VARY — the advantage estimator only. PPO and GRPO are run on one and the same stack, so the estimator is the single difference between the two arms.

FIX — model checkpoint, dataset slice, decoding path, reward grader, framework, schedule and seed set.

Measured outcome. Paired $\Delta = -0.002$, $p = 0.374$: the estimator effect is small and not statistically significant.

Pillar 2 · signal starvation

VARY — the internal signal state of the batch: per-step Zero-Variance Fraction and its collapse decomposition, logged at every step rather than summarised after the run.

FIX — the training stack is untouched; the diagnostic observes, it does not intervene.

Measured outcome. 0.16 mean ZVF on Qwen3-8B / GSM8K at $G = 8$. All nine rows of the method panel, vanilla GRPO included, are a declared simulation projection.

Pillar 3 · group size

VARY — the group size $G \in \{2, 4, 8, 16\}$ across seeds, holding model, data, reward and schedule fixed.

FIX — model, data, reward grader, learning-rate schedule and evaluation split.

Measured outcome. Mean ZVF falls $0.838 \rightarrow 0.631$ ($G = 2 \rightarrow 16$); the training-reward ratio to $G = 2$ is ≈ 1.00 , equivalent up to $G = 8$ but not established at $G = 16$.

Pillar 4 · held-out behaviour

VARY — the method, Dr. GRPO versus GRPO, compared pre-RL against post-RL on data the training loop never saw.

FIX — the pre-RL checkpoint and the held-out split, evaluated side by side under frozen sampling.

Measured outcome. Held-out gain small and not significant; length drifts $194.4 \rightarrow 183.5$ tokens under the cap.

Vary exactly one factor; hold every other factor fixed.

Headline configuration (varies by study; Table 4.2): LoRA rank 32; Adam with $\beta_1 = 0.9$, $\beta_2 = 0.95$; learning rate 1×10^{-4} ; seeds {42, 123, 456, 789, 1024}; one shared reward grader and one shared decoding path across back-ends; Welch / Mann–Whitney, Cohen’s d with Hedges–Olkin intervals, 95% bootstrap CIs (10,000 resamples) and a declared multiple-comparison family, declared but not pre-registered.